

# CareerCompass AI

## Your Personal Career Navigator

### *Project Proposal Document*

## 1. Project Overview

**CareerCompass AI** is a mobile application designed to guide users through their professional journey by providing **personalized career recommendations**, **skill development tracking**, and **AI-powered job market insights**. Built using **object-oriented programming (OOP)** principles, the application combines modern technologies with an intuitive user experience to help job seekers, students, and professionals make informed career decisions.

## 2. Problem Statement

Navigating today's job market is challenging due to:

- Information overload from multiple platforms
- Lack of personalized career guidance
- Difficulty identifying relevant skills and certifications
- No centralized tool to track career progress
- Limited access to mentorship and mock interview opportunities

Existing solutions are fragmented, often requiring users to switch between apps for resumes, job searches, skill development, and networking.

## 3. Proposed Solution

**CareerCompass AI** solves these problems by offering an **all-in-one career guidance platform** that includes:

- **AI-powered resume analysis** with actionable feedback
- **Personalized job matching** based on skills, experience, and preferences

- **Skill development tracking** with gamified challenges and certification recommendations
- **Mock interview simulator** with real-time feedback
- **Mentorship connection system**
- **Project portfolio builder** to showcase user work
- **Real-time job market trends** using public APIs

## 4. Innovation Highlights

| Innovation                           | Description                                                                                                                    |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>AI Career Assistant</b>           | Built-in NLP engine analyzes resumes and provides personalized recommendations for keyword optimization and ATS compatibility. |
| <b>Gamified Skill Development</b>    | Users earn badges, points, and levels by completing career-related tasks, maintaining high engagement and motivation.          |
| <b>Unified Career Dashboard</b>      | Single view showing resume score, job matches, skill progress, upcoming interviews, and mentorship sessions.                   |
| <b>Smart Notifications</b>           | Context-aware alerts using OOP design patterns (Observer, Strategy) to notify users of relevant job postings and skill trends. |
| <b>Portfolio Builder Integration</b> | Automatically generates shareable portfolios with GitHub links, project descriptions, and demo videos.                         |
| <b>Modular OOP Architecture</b>      | Clear separation of concerns using inheritance, polymorphism, and encapsulation for scalability and maintainability.           |

Table 1: Key Project Innovations

## 5. Core Features (MVP)

### 5.1 User Management

- User registration and login (email, Google, LinkedIn)
- Profile creation with education, experience, skills, and career goals

### 5.2 Resume Analyzer

- Upload resume (PDF/DOCX)
- AI analysis: keyword extraction, formatting suggestions, ATS score
- Improvement tips with before/after examples

### **5.3 Job Matching**

- Match jobs from public APIs based on user profile
- Save and apply to jobs directly within the app

### **5.4 Skill Development**

- Skill assessment quizzes
- Recommended courses and certifications
- Progress tracking with visual charts

### **5.5 Mock Interview**

- Role-specific interview questions
- Voice input and AI feedback on responses
- Peer-to-peer mock interview scheduling

### **5.6 Mentorship**

- Browse and connect with mentors by industry
- Schedule video/chat sessions
- Rating and review system

### **5.7 Portfolio Builder**

- Step-by-step project suggestions
- GitHub integration
- Portfolio export as a web page or PDF

## **6. Technology Stack**

## **7. Object-Oriented Design**

The application will be structured using core OOP principles:

| Component      | Technology                                                                  |
|----------------|-----------------------------------------------------------------------------|
| Language       | Java / Kotlin (Android) or Swift (iOS) – OOP-focused                        |
| Architecture   | MVVM with Clean Architecture principles                                     |
| Backend        | Firebase (Authentication, Firestore, Cloud Functions) or custom Node.js API |
| AI/NLP         | Google Cloud Natural Language API or OpenAI API                             |
| APIs           | LinkedIn API, Indeed API, Glassdoor API                                     |
| Database       | Firebase Firestore or PostgreSQL                                            |
| Authentication | Firebase Auth / OAuth 2.0                                                   |
| Notifications  | Firebase Cloud Messaging                                                    |

Table 2: Technology Stack

| OOP Concept   | Application                                                                                                                                                     |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Encapsulation | User data, resume content, and progress are encapsulated within classes with private fields and public methods.                                                 |
| Inheritance   | Base <code>User</code> class extended by <code>JobSeeker</code> , <code>Mentor</code> , <code>Employer</code> subclasses.                                       |
| Polymorphism  | Different notification types ( <code>JobAlert</code> , <code>Reminder</code> , <code>Tip</code> ) handled through a common <code>Notification</code> interface. |
| Abstraction   | API services abstracted behind interfaces for easy testing and swapping.                                                                                        |

Table 3: OOP Concepts Application

## Sample Class Structure

```
// Abstract User class
public abstract class User {
    private String id;
    private String name;
    private String email;

    public abstract void updateProfile();
}

// JobSeeker subclass
public class JobSeeker extends User {
    private Resume resume;
    private List<Skill> skills;
    private List<Job> savedJobs;

    public void applyToJob(Job job) { ... }
    public double getResumeScore() { ... }
}

// Resume class
public class Resume {
    private String content;
    private double atsScore;
```

```

    public AnalysisResult analyze() { ... }
    public List<String> suggestImprovements() { ... }
}

```

## 8. Development Roadmap

| Phase                        | Duration | Deliverables                                                  |
|------------------------------|----------|---------------------------------------------------------------|
| Phase 1: Planning & Design   | 2 weeks  | Requirements, UML diagrams, wireframes, database schema       |
| Phase 2: Core OOP Classes    | 3 weeks  | Implement base classes, authentication, profile management    |
| Phase 3: Feature Development | 6 weeks  | Resume analyzer, job matching, skill tracking, mock interview |
| Phase 4: AI Integration      | 2 weeks  | Integrate NLP APIs, fine-tune recommendations                 |
| Phase 5: Testing             | 2 weeks  | Unit tests, UI tests, beta user feedback                      |
| Phase 6: Deployment          | 1 week   | App store submission, documentation                           |

Table 4: Development Roadmap

## 9. Resources Needed

| Resource                | Details                                                                      |
|-------------------------|------------------------------------------------------------------------------|
| <b>Development Team</b> | 1–2 mobile developers (OOP expertise), 1 UI/UX designer, 1 backend developer |
| <b>Tools</b>            | Android Studio / Xcode, Figma, Git/GitHub, Firebase Console                  |
| <b>APIs</b>             | OpenAI/Google Cloud credits, LinkedIn/Indeed API access                      |
| <b>Budget</b>           | Server costs, API usage fees, App Store developer accounts (\$100/year)      |

Table 5: Resources Needed

## 10. Success Metrics

## 11. Conclusion

**CareerCompass AI** leverages **object-oriented programming** to deliver a scalable, maintainable, and innovative mobile application that addresses real-world career guidance challenges. By combining AI, gamification, and personalized recommendations, the app

| <b>Metric</b>                  | <b>Target</b> |
|--------------------------------|---------------|
| User sign-ups (first 3 months) | 5,000         |
| Resume analyses completed      | 2,000         |
| Job matches saved              | 1,500         |
| Skill challenges completed     | 3,000         |
| Mock interviews conducted      | 500           |
| Mentor connections made        | 200           |
| User retention rate (monthly)  | 40%           |

Table 6: Success Metrics

empowers users to take control of their professional development and succeed in the job market.

— **End of Document** —